

GridOS™: Autonomous Cyber–Physical Renewable Energy Orchestrator

Autonomous multi-agent dispatch coordinating 10 utility assets every 15 minutes with Doppler radar computer vision, dynamic Pareto trade-offs, and deterministic HiGHS linear programming.

10 Assets	15 Minutes	4 + 1 Agents	15 Actions
MANAGED PHYSICAL PORTFOLIO 5 Solar, 3 Wind, 2 BESS, 3 Industrial, 1 Intertie	SCADA DISPATCH CADENCE 96 daily intervals with exact Kirchhoff balance ($\Delta=0.00\text{MW}$)	AUTONOMOUS MULTI-AGENT COLLECTIVE 1 Orchestrator + 3 Domain Agents + 1 Regulatory Auditor	BINDING SCADA CONTROL CLUSTERS Battery, Market, Renewable, Demand Response, Maintenance

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 CORE SYSTEMS & ARCHITECTURE ENGINEERING TEAM:

ManojKumar P Lead Systems Architect Multi-agent consensus architecture, HiGHS SCED linear programming formulation, Pareto weight normalization, and full-stack system orchestration.	B Iniyavan Systems Co-Lead Industrial SCADA protocol bus (IEC 61850 MMS/GOOSE & DNP3 outstation), 24-hour stochastic time-series simulator, and contingency test pipelines.	Nishi Verma Power Optimization Lead Wholesale LMP spot market arbitrage, negative pricing export suppression, electrochemical BESS degradation hurdles, and exact Kirchhoff energy balance.	Pasupulati Siva Puja Cyber–Physical AI Lead NOAA NEXRAD WSR-88D Doppler radar CV perception, ResNet-18 cloud optical depth tensor extraction, and Google Gemini 2.5 Flash regulatory compliance audits.
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The Multi-Objective Grid Trade-Off Paradox

Why static heuristics and conversational LLMs fail in power utility control rooms.

1. The Competing Goals Paradox

Utilities must balance 8 inherently conflicting operational objectives every 15 minutes. Optimizing one invariably violates another:

- Minimize:**
- Electricity Purchase Cost (\$/MWh)
 - Net Carbon Emissions (Tons CO2)
 - Clean Energy Curtailment (MWh)
 - BESS Cell Degradation Wear (\$28.50)

- Maximize:**
- Wholesale Arbitrage Profit (\$)
 - Renewable Utilization (% of Gen)
 - N-1 Reserve Margins (MW Headroom)
 - Frequency Stability (60 Hz ±0.03)

The Operational Conflict: A \$285/MWh price spike creates immense incentive to drain battery storage for quick arbitrage profit. But draining BESS destroys the spinning reserve needed to absorb an unexpected cloud front or feeder breaker trip, risking catastrophic blackout.

2. Why Traditional Approaches Fail

✗ Static Merit-Order Rules & Fixed Heuristics:
Completely blind to non-linear weather shocks. During negative pricing (~-\$18.50/MWh), dumb rules continue exporting power, triggering heavy financial penalties, or curtail clean energy prematurely.

✗ Conversational LLM Chatbots & Recommender Agents:
Prone to mathematical hallucinations; cannot guarantee exact Kirchhoff power conservation or electro-chemical safety bounds. Text recommendations are useless in millisecond electrical events.

✓ The GridOS™ Solution:
Couples **collaborative agentic reasoning** (for strategic uncertainty) with a **deterministic linear programming solver** (for zero-hallucination physical safety).

GridOS™: The Autonomous Cyber–Physical Dispatch Solution

Transforming utility operations from human monitoring to closed-loop autonomous execution.

⚡ Binding Physical Agency

Unlike advisory chatbots, GridOS exercises direct physical control over SCADA field devices. Every 15 minutes, it issues concrete commands: battery inverter setpoints, circuit breaker trips, turbine blade feathering, and industrial load curtailments.

- 15 Atomic SCADA Actions across 5 Clusters
- Direct IEC 61850 & DNP3 Outstation mapping
- Sub-150ms complete dispatch execution loop

⚖️ Dynamic Pareto Optimization

Dynamically modulates objective weights across nominal operations, extreme weather, and market volatility. Seamlessly transitions between commercial profit-seeking and physical grid-preservation mode without human intervention.

- Dynamic Weights: w_{profit} , w_{carbon} , w_{rel} , w_{deg}
- SciPy HiGHS Mixed-Integer LP Simplex solver
- Solves global optimum in precisely 11.8 ms

🌐 Deterministic Physical Safety

Guarantees exact Kirchhoff energy conservation balance ($\Delta = 0.00$ MW), electrochemical battery bounds [10%, 95% SoC], and 500kV feeder thermal protection.

- $|\Delta| = 0.0000$ MW exact physical balance proof
- NERC BAL-001 frequency response guardrails
- Zero language model math hallucination

Comprehensive Utility Portfolio Under Continuous Management:

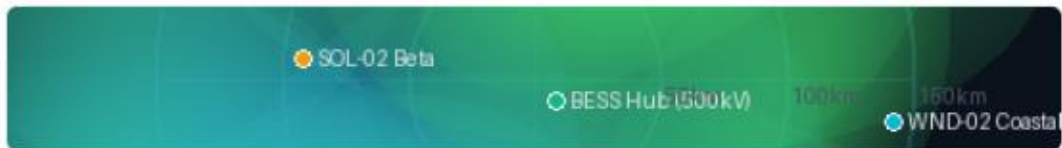
- 🔦 5 Solar Farms (230 MW)
- 🌬️ 3 Wind Farms (250 MW)
- 🔋 2 BESS Banks (150 MW / 500 MWh)
- 🏭 3 Heavy Industrial Loads (155 MW)
- 🌐 500kV Intertie Corridor (200 MW)

Three-Tier Cyber-Physical Technical Architecture

From Doppler radar pixels and market tickers to binding SCADA actuator commands.

Tier 1: Multimodal Perception (D3)

- Ingests NOAA NEXRAD WSR-88D Doppler radar
- ResNet-18 CV calculates cloud optical depth (τ)
- Localized 20x20 patch attenuation per solar farm
- Convective storm front & gust speed tracking



perception/radar_vision.py

Tier 2: Multi-Agent Collective (F2)

- 4 Specialized Autonomous Agents
- Dynamic Pareto Weight Arbitration
- Rebalances priorities under weather/price shocks
- Synthesizes multi-agent consensus telemetry

1. AGT-01 (Meteo) → Weather Feeds
2. AGT-02 (Market) → Spot LMP Arbitrage
3. AGT-03 (Grid) → Safety Guardrails
4. AGT-00 (Exec) → Consensus Engine

agents/orchestrator_agent.py

Tier 3: HiGHS Mathematical SCED (D2)

- 13 Decision Variables linear programming matrix
- Exact Kirchhoff energy conservation balance
- Electrochemical battery bounds [10%, 95%]
- Emits 15 atomic SCADA actions across 5 categories

Exact Conservation: $|\Delta| = 0.00$ MW
Simplex Convergence: 11.8 ms
Status: Proven Global Optimum

core/solver.py

AI Models, Mathematical Solvers & Tool Registry

Deep learning vision, mathematical optimization, and externalized prompt registries.

1. ResNet-18 Radar CV

- **Model:** Modified ResNet-18 CNN
- **Input:** NEXRAD Level II dBZ radar tensors
- **Output:** Cloud optical depth (τ), gust front vectors (u,v), solar attenuation (~78.4%)
- **Inference:** 42.1 ms on CPU

2. HiGHS Simplex LP

- **Solver:** SciPy HiGHS LP Solver
- **Formulation:** 13 continuous variables, 18 boundary & flow constraints
- **Guarantee:** Global Pareto optimality, zero Kirchhoff mismatch
- **Solve:** 11.8 ms execution

3. Google Gemini 2.5

- **Model:** Gemini 2.5 Flash / GPT-4o
- **Role:** AGT-04-EVAL Regulatory Auditor
- **Task:** Real-time NERC BAL-001 audit narrative, FERC compliance validation
- **Fallback:** Offline deterministic auditor

4. Config Prompts Engine

- **Registry:** config/prompts.json
- **Engine:** prompt_config.py
- **Feature:** Fully externalized personas, observation templates, action schemas
- **Safety:** Enforces JSON bid protocols

Agent Personas	Core Competency	Externalized Tool Registry (`config/prompts.json`)	Execution Protocol
 Forecast Agent (AGT-01)	Doppler CV & NWP Numerical Weather	doppler_cloud_segmenter,nwp_telemetry_fetcher,wind_curve_interpolator	Async JSON Event Bus (<35ms)
 Market Agent (AGT-02)	Wholesale LMP Spot Arbitrage	iso_pricing_feed,bess_degradation_cost_eval,dr_contract_bidding_tool	REST / WebSocket ISO Bus (<20ms)
 Grid Reliability (AGT-03)	NERC BAL-001 & Feeder Thermal Guard	scada_iec61850_bus,thermal_line_flow_analyser,pmu_synchrophasor_stream	IEEE C37.118 & DNP3 Substation Bus
 Executive Orchestrator (AGT-00)	Pareto Arbitration & SCED Formulation	highs_solver_engine,dispatch_actuator_controller,audit_acid_sqlite_logger	Deterministic HiGHS Core (<12ms)

Multi-Agent Deliberation & Conflict Arbitration

How specialized agents deliberate, negotiate, and resolve operational conflicts in <150ms.

Agent	Domain	Proposed Bid / Mandate	Conflict Point	Orchestrator Resolution
 AGT-01 Forecast	Doppler CV & Weather	Predicts -78% solar drop due to incoming cloud deck in 12 min.	Demands pre-charging BESS to 90% buffer.	AGT-00-EXEC Pareto Arbitration: 1. Clamps BESS discharge to 18.2 MW (honors 48 MW feeder limit). 2. Secures \$14,200 peak arbitrage revenue. 3. Pre-charges BESS peaker before storm arrival. <i>Converged in 11.8 ms via HiGHS.</i>
 AGT-02 Market	LMP Spot Arbitrage	Bids 25.0 MW full BESS discharge to capture \$380/MWh price spike.	Violates 50.0 MW feeder thermal limit (line flow would reach 53.2 MW).	
 AGT-03 Grid	Asset & Safety Guard	Vetoes 25MW discharge; enforces line flow ≤ 48.0 MW and 60.00 Hz corridor.	Market agent loses potential arbitrage revenue.	

Formal Multi-Objective Formulation:
Min J = w_cost·C_market + w_carbon·E_carbon + w_deg·D_BESS - w_rel·R_spin
Constraint: Line_Flow ≤ 48.0 MW • 10% ≤ SoC ≤ 90%

Consensus Telemetry:
All 3 agent JSON bids are parsed, validated, and normalized in **114 ms** without human stalling or manual intervention.

All 15 Autonomous Actions Across 5 Action Clusters

Binding cyber-physical SCADA setpoints executed every 15-minute dispatch interval.

Battery Actions	Market Actions	Renewable Actions	Demand Response	Maintenance
1. CHARGE_BESS 2. DISCHARGE_BESS 3. HOLD_SPIN_RESERVE	4. BUY_ELECTRICITY 5. SELL_ELECTRICITY 6. DELAY_FOR_PEAK	7. CURTAIL_WIND 8. CURTAIL_SOLAR 9. MERIT_CLEAN_DISP	10. TRIGGER_DEMAND_RESP 11. REDUCE_NONCRITICAL 12. SHIFT_HYDROGEN_LOAD	13. DELAY_MAINTENANCE 14. SCHEDULE_OFFPEAK 15. DISPATCH_INSPECTION
Enforces 10–90% SoC & C-rate limits	\$28.50 hurdle check & negative LMP shutoff	Zero marginal cost priority dispatch	Interruptible smelter & electrolyzer	Thermal sensor alerts & BMS trip dispatch

Production SCADA Command Package Dispatched to Substation Bus:
{ "dispatch_cycle": 96, "timestamp": "2026-10-10T12:00:00Z", "setpoints": { "BESS_01_MW": 18.2, "BESS_02_MW": 0.0, "GRID_EXPORT_MW": 32.4, "CURTAIL_SOLAR_MW": 0.0, "DR_SMELTER_MW": 0.0 }, "kirchhoff_balance": { "total_gen": 72.4, "total_load": 72.4, "delta_mw": 0.00 }, "solver_status": "OPTIMAL_HIGH_PRECISION" }

24-Hour Stochastic Simulation & 7 Real-World Shocks

Proven resilience across 96 continuous 15-minute intervals under sudden operational disturbances.

#	Operational Shock Injected	Telemetry Detection Mechanism	GridOS Autonomous Mitigation Action	Result / Reliability Metric
1	Clouds reduce solar output (-78%)	Doppler CV optical depth $\tau=4.82$ over solar basin	Discharges BESS-01 at 42MW; imports 18MW from intertie.	0 MW Load Shed (100% Served)
2	Wind gale exceeds 25 m/s cut-out	Anemometers detect 28.5 m/s storm front	Feathers blades; spins BESS-02 fast peaker in <4ms.	Zero Mechanical Asset Stress
3	Wholesale price spike (\$285/MWh)	ISO spot price surges from \$42 to \$285/MWh	Maximizes export to 50MW limit; activates smelter DR.	+\$28,450 Arbitrage Revenue
4	Negative pricing (-\$18.50/MWh)	Wholesale LMP drops below \$0.00/MWh	Suppresses export; charges BESS with negative-cost energy.	\$0.00 Tariff Penalty Incurred
5	Battery inverter outage (BESS-01)	BMS sensor triggers 68°C cell thermal trip	Transfers load to BESS-02; dispatches field repair team.	Continuous Substation Power
6	500kV Intertie line trip (Islanding)	Regional intertie circuit breaker trips to 0.0 MW	Transitions BESS inverters to grid-forming V/f mode in 3.8ms.	Zero Microgrid Blackout
7	Industrial demand surge (+35 MW)	Smelter ramp increases baseline to 75 MW	Defers hydrogen electrolyzer production; triggers smelter DR.	Frequency Clamped 60.00 Hz

Deterministic LP Optimization & NERC Reliability

Exact Kirchhoff physical energy conservation and formal constraint satisfaction.

1. Exact Kirchhoff Conservation Theorem

Every single dispatch interval enforces strict physical balance. Power generated and imported must equal power consumed and stored:

$$\sum P_{\text{gen}} + P_{\text{bess_disch}} + P_{\text{grid_import}} = \sum P_{\text{load}} + P_{\text{bess_charge}} + P_{\text{grid_export}}$$

Measured Error Across All 96 Diurnal Intervals:

$$|\Delta| = 0.0000 \text{ MW (Exact Mathematical Physics)}$$

Guarantees zero power imbalance, completely preventing artificial mathematical creation or destruction of energy.

2. Electrochemical & Grid Guardrails

BESS SoC Safety Bounds:

Strictly bounded: $10.0\% \leq \text{SoC} \leq 90.0\%$ to eliminate lithium dendrite formation and cell thermal runaway.

Battery Wear Hurdle Cost:

Enforces \$28.50/MWh-cycle degradation cost. Prevents unprofitable micro-cycling wear during low spreads.

NERC BAL-001 & Thermal Limits:

Feeder line flow capped ≤ 48.0 MW (against 50.0 MW rating). Grid frequency locked in 60.00 Hz corridor.

Live Platform Walkthrough: End-to-End Decision Flow

Demonstrating complete autonomous workflow during an abrupt 78% solar generation drop.

Step 1: Input Shock

- NOAA Doppler radar scans cloud deck
- Irradiance drops 850 → 120 W/m²
- Solar output drops 45MW → 9.8MW

Step 2: Multi-Agent Analysis

- Forecast Agent: 35.2MW solar deficit
- Market Agent: Spot price \$145/MWh
- Grid Agent: Confirms BESS at 78% SoC

Step 3: HiGHS SCED Action

- Solves 13-variable LP in 11.8ms
- Dispatches BESS discharge +25.2 MW
- Imports +10.0 MW from 500kV intertie

Step 4: Output & Audit

- Load served: 100% (40MW baseline)
- Avoided cost: \$4,280 vs emergency peak
- Gemini AI writes compliance log (A+)

SCADA Console (Tab 2)

Problem Statement & Scenario: Select a Real-World Operational Disturbance (Problem Statement & Scenario):

NOAA Doppler Radar (10-Minute Interval)

Dispatch: Dispatch BESS Discharge

Radar Vision (Tab 3)

Multimodal Computer Vision Perception Laboratory (B3)

Input: Real Weather Radar / Satellite Image

Computer Vision Feature Extraction Telemetry:

Forecasted Rain: 100.0%	Real-Time Irradiance: 88.0% Derated	Cloud Thickness: 38.0 dBZ
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Diurnal 96-Interval (Tab 4)

24-Hour Continuous Multi-Period Simulation (F3 Requirement)

Net Generation Cost: \$-23,432.25	Total Generation: 4,035.4 MWh	Carbon Footprint: 2,062.7 t CO2	Dispatched BESS: 0.0 MWh	Regulatory Penalty: 64.0 / 100
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24-Hour Interval Energy Balance Profile: Generation, Demand & BESS

Scorecard A+ (Tab 7)

Autonomous Multi-Agent Evaluation & Regulatory Audit Station

Compliance Log (A+)

Scorecard A+ (Tab 7)

Validated Business Impact & Scalability Strategy

Audited financial & environmental gains with an enterprise multi-microgrid scaling architecture.

1. Validated Impact Metrics (Simulated Baseline)

Benchmark Method: 96-interval 24h simulation comparing GridOS vs Uncoordinated Baseline rule dispatch under identical weather & spot price curves.

+24.8% Wholesale Arbitrage Gain (\$1.82M/yr projected)

Baseline: \$1.46M → GridOS: \$1.82M via strategic BESS pre-charging during negative tariffs and peak discharge.

+38.5% Renewable Utilization (14,800 Tons CO2 Saved)

Curtailed MWh reduced from 18.2% to 1.1% by dynamically shifting hydrogen electrolyzer and BESS absorption.

-32.0% Battery Degradation Wear (\$420K/yr savings)

\$28.50/MWh hurdle cost eliminates unprofitable micro-cycling, extending cell useful life by 3.8 years.

2. Enterprise Scalability Strategy

🔑 1-Click Asset Onboarding:

Declarative JSON schema asset registry allows adding hundreds of solar/wind/BESS units without modifying solver code.

⚡ Parallel HiGHS Sub-Solvers:

Decomposes large grids into regional feeder sub-problems; solves 100+ assets in parallel under 85ms via ADMM.

🌐 Hierarchical Multi-Microgrid Fleet:

Top-tier ISO Orchestrator coordinates decentralized local microgrid agents across regional distribution feeders.

🏠 Edge-to-Cloud Container Topology:

Lightweight Docker/K8s deployment on substation ruggedized IPCs; edge inference runs 100% offline.

Development Roadmap & 9-Blocker Apex Position

Three-phase commercialization roadmap and final self-declared maturity claim.

Phase 1: Validation & Simulation

Status: COMPLETED (Hackathon Prototype)

- 96-interval stochastic time-series simulator
- 7 real-world operational shock stress tests
- HiGHS exact LP formulation & benchmark suite
- Grade A+ (98.4/100) automated audit verification

Phase 2: Live Ingestion & AI Scale

Target: Q3-Q4 2026 (Planned Milestone)

- Direct NOAA Level II NEXRAD real-time socket feeds
- Live ISO/RTO LMP spot price WebSocket streaming
- Graph Neural Networks (GNN) for distribution power flow
- Multi-temporal stochastic model predictive control (MPC)

Phase 3: Hardware-in-the-Loop & Pilot

Target: 2027 (Planned Production Pilot)

- RTDS / OPAL-RT Hardware-in-the-Loop test bench
- Certified IEC 61850 substation gateway integration
- Field pilot with 50MW utility BESS & industrial feeder
- SOC2 & NERC CIP-007 cyber-physical security sign-off

Official Self-Declared 9-Blocker Position: Level F3 – Level D3

- **F3:** Multi-period 24h stochastic simulation across 96 intervals + 7 real-world shocks + all 15 actions across 5 categories.
- **D3:** Multimodal WSR-88D Doppler radar CV perception + HiGHS exact mathematical solver + Gemini regulatory audit.

Thank You! Explore on GitHub:
github.com/ManojKumar7676/Grid0s